

Ruofan Wu

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I am a third-year PhD candidate in CSE at the University of Michigan, advised by Prof. Mosharaf Chowdhury. My research focuses on energy-efficient AI systems and compilers. I aim to reduce energy consumption while maximizing performance under power constraints. Because energy efficiency and performance are tightly coupled, my work explores their joint optimization, with a particular emphasis on GPU kernel scheduling.

EDUCATION

University of Michigan <i>Ph.D. candidate in Computer Science and Engineering</i>	Ann Arbor, MI, USA <i>Sep 2024 - Present</i>
Renmin University of China <i>M.E. in Computer Application Technology</i>	Beijing, China <i>Sep 2021 - Jun 2024</i>
Renmin University of China <i>B.E. in Data Science and Big Data Technology</i>	Beijing, China <i>Sep 2017 - Jun 2021</i>

EXPERIENCE

NVIDIA <i>PhD Research Intern at ADLR & Compute Arch</i> <i>Mentored by Dr. Ahmet Inci, Dr. Bhargava Gopireddy, and Dr. Deepak Narayanan</i>	Santa Clara, CA, USA <i>May 2026 - Aug 2026</i>
Microsoft Software Technology Center Asia <i>Research Intern at Bing, mentored by Dr. Zhen Zheng</i>	Beijing, China <i>Oct 2023 - Jun 2024</i>
Alibaba Cloud <i>Research Intern at Platform of Artificial Intelligence, mentored by Dr. Zhen Zheng</i>	Hangzhou, China <i>Jan 2022 - Sep 2023</i>
Microsoft Research Asia <i>Research Intern at Systems Research Group, mentored by Dr. Jilong Xue and Dr. Fan Yang</i>	Beijing, China <i>Apr 2021 - Dec 2021</i>
North Carolina State University <i>Collaborator with PICTURE Research Group, mentored by Prof. Xipeng Shen</i>	Remote <i>Nov 2019 - Mar 2021</i>
Renmin University of China <i>Student at Database and Intelligent Information Retrieval Group, advised by Prof. Feng Zhang</i>	Beijing, China <i>Oct 2019 - Jun 2024</i>

PUBLICATIONS

- **Energy calculus: a compositional algebra of energy in computational systems**, Mosharaf Chowdhury, Jae-Won Chung, Jeff J. Ma, Nishil Talati, **Ruofan Wu**. Preprint, 2026.
- **Where do the joules go? Diagnosing inference energy consumption**, Jae-Won Chung, **Ruofan Wu**, Jeff J. Ma, Mosharaf Chowdhury. Preprint, 2026.
- **Kareus: joint reduction of dynamic and static energy in large model training**, **Ruofan Wu**, Jae-Won Chung, Mosharaf Chowdhury. **OSDI**, 2026.
- **TetriServe: efficient DiT serving for heterogeneous image generation**, Runyu Lu*, Shiqi He*, Wenxuan Tan, Shenggui Li, **Ruofan Wu**, Jeff J. Ma, Ang Chen, Mosharaf Chowdhury. **ASPLOS**, 2026.
- **The ML.ENERGY benchmark: toward automated inference energy measurement and optimization**, Jae-Won Chung, Jeff J. Ma, **Ruofan Wu**, Jiachen Liu, Oh Jun Kweon, Yuxuan Xia, Zhiyu Wu, Mosharaf Chowdhury. **NeurIPS Datasets and Benchmarks (Spotlight)**, 2025.
- **PluS: highly efficient and expandable ML compiler with pluggable graph schedules**, **Ruofan Wu**, Zhen Zheng, Feng Zhang, Chuanjie Liu, Zaifeng Pan, Jidong Zhai, Xiaoyong Du. **USENIX ATC**, 2025.
- **RecFlex: enabling feature heterogeneity-aware optimization for deep recommendation models with flexible schedules**, Zaifeng Pan, Zhen Zheng, Feng Zhang, Bing Xie, **Ruofan Wu**, Shaden Smith, Chuanjie Liu, Olatunji Ruwase, Xiaoyong Du, Yufei Ding. **SC**, 2024.

PUBLICATIONS (CONTINUED)

- **RECom: a compiler approach to accelerating recommendation model inference with massive embedding columns**, Zaifeng Pan, Zhen Zheng, Feng Zhang, **Ruofan Wu**, Hao Liang, Dalin Wang, Xiafei Qiu, Junjie Bai, Wei Lin, Xiaoyong Du. **ASPLOS**, 2024.
- **Expanding the edge: enabling efficient Winograd CNN inference with deep reuse on edge device**, Feng Zhang, **Ruofan Wu**, Jiawei Guan, Zhen Zheng, Xiaoguang Guo, Xiao Zhang, Xiaoyong Du, Xipeng Shen. **IEEE TKDE**, 2023.
- **ROLLER: fast and efficient tensor compilation for deep learning**, Hongyu Zhu, **Ruofan Wu**, Yijia Diao, Shanbin Ke, Haoyu Li, Chen Zhang, Jilong Xue, Lingxiao Ma, Yuqing Xia, Wei Cui, Fan Yang, Mao Yang, Lidong Zhou, Asaf Cidon, Gennady Pekhimenko. **OSDI**, 2022.
- **TREC: transient redundancy elimination-based convolution**, Jiawei Guan, Feng Zhang, Jiesong Liu, Hsin-Hsuan Sung, **Ruofan Wu**, Xiaoyong Du, Xipeng Shen. **NeurIPS**, 2022.
- **DREW: efficient Winograd CNN inference with deep reuse**, **Ruofan Wu**, Feng Zhang, Jia Wei Guan, Zhen Zheng, Xiaoyong Du, Xipeng Shen. **TheWebConf**, 2022.
- **YuenyeungSpTRSV: a thread-level and warp-level fusion synchronization-free sparse triangular solve**, Jiya Su, Feng Zhang, Weifeng Liu, Bingsheng He, **Ruofan Wu**, Xiaoyong Du, Rujia Wang. **IEEE TPDS**, 2021.
- **CapelliniSpTRSV: a thread-level synchronization-free sparse triangular solve on GPUs**, Jiya Su, Feng Zhang, Weifeng Liu, Bingsheng He, **Ruofan Wu**, Xiaoyong Du, Rujia Wang. **ICPP**, 2020.

ADDITIONAL INFORMATION

- **Talk:** NeurIPS 2025 tutorial on **Energy and Power as First-Class ML Design Metrics**
- **Academic Service:** Reviewer for IEEE TPDS (Dec 2023)
- **Teaching Assistant:** Parallel Architecture and Programming (Renmin University of China, Fall 2022)
- **Honors and Awards:** Renmin University of China: Excellent Bachelor Dissertation; Scholarship (2019–2023); Sa-Shixuan Fund Scholarship (2019)
- **Programming Languages:** Python, C/C++/CUDA
- **Tools and Frameworks:** Megatron-LM, TransformerEngine, PyTorch (Dynamo), Nsight, Zeus, NVML